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R-Tuning: Instructing Large Language Models to Say 'I Don't Know'

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Abstract

Large language models (LLMs) have revolutionized numerous domains with their impressive performance but still face their challenges. A predominant issue is the propensity for these models to generate non-existent facts, a concern termed *hallucination*. Our research is motivated by the observation that previous instruction tuning methods force the model to complete a sentence no matter whether the model knows the knowledge or not. When the question is out of the parametric knowledge, it will try to make up something and fail to indicate when it lacks knowledge. In this paper, we present a new approach called **R**efusal-Aware Instruction **Tuning** (**R-Tuning**). This approach is formalized by first identifying the disparity in knowledge encompassed by pre-trained parameters compared to that of instruction tuning data. Then, we construct the refusal-aware data based on the knowledge intersection, to tune LLMs to refrain from responding to questions beyond its parametric knowledge. Experimental results demonstrate R-Tuning effectively improves a model's ability to answer known questions and refrain from answering unknown questions. Furthermore, when tested on out-of-domain datasets, the refusal ability was found to be a meta-skill that could be generalized to other tasks. Further analysis surprisingly finds that learning the uncertainty results in better calibration and an improved ability to estimate the uncertainty than uncertainty-based testing.[1]

1 Introduction

Large language models (LLMs) have demonstrated remarkable performance across numerous tasks; however, they are also plagued by various issues, such as the propensity of large models to fabricate non-existent facts, a phenomenon commonly referred to as *hallucination* (Maynez et al., 2020a).

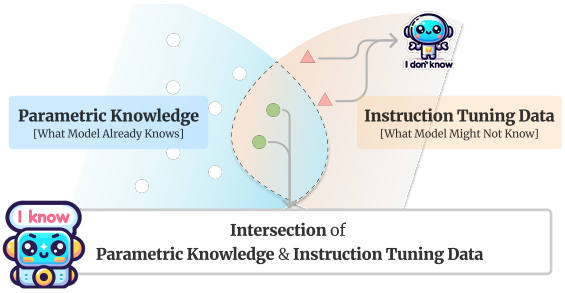


Figure 1: An illustration of the parametric knowledge distribution and the instruction tuning data distribution. Pre-training embeds a large volume of parametric knowledge, while fine-tuning may involve knowledge that is not necessarily in the parametric knowledge. We explore the benefits of differentiating instruction tuning data based on parametric knowledge.

Towards mitigating the hallucination, current mainstream approaches include retrieval-based methods (Peng et al., 2023; Li et al., 2023b; Luo et al., 2023), verification-based methods (Manakul et al., 2023; Elaraby et al., 2023; Cohen et al., 2023; Du et al., 2023; Gou et al., 2023), and so forth.

In this paper, we first identify the cause of hallucination, attributing it to the significant gap existing between the knowledge derived from human-labeled instruction tuning datasets and the parametric knowledge of LLMs. In the process of developing a large model, previous studies (Min et al., 2022; Wang et al., 2023; Zhou et al., 2023) demonstrate that almost all knowledge is acquired in the pre-training stage, while instruction tuning teaches formatting and chain-of-thought prompting guides knowledge elicitation. Consider Figure 1 as an example. During pre-training, models embed a large volume of factual knowledge, compressing it within their parameters and the fine-tuning process may include data that is out of the parametric knowledge. However, traditional fine-tuning methods force the models to complete each sentence. Even when faced with questions beyond their knowledge boundary, they venture to provide

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[1]Our code is available at https://github.com/shizhediao/R-Tuning.

an answer. Training a model exclusively on correct answers inadvertently teaches it to guess rather than admit its ignorance. Consequently, if we never train the model to articulate *"I don't know"* as a response, it remains unequipped to do so when confronted with unknowns. Addressing this challenge, we assert that enabling a model to astutely respond based on its own knowledge limit is of paramount importance. This motivates us to tune our model on the intersection of parametric knowledge and the instruction tuning data, leading to a model expressing its confidence value and refusing to answer unknown questions.

In light of this, we propose a novel instruction tuning method, **R**efusal-Aware Instruction **Tuning** (**R-Tuning**). R-Tuning aims to endow the model with refusal-aware answering ability by recognizing when they should — and shouldn't — claim knowledge. Specifically, R-Tuning introduces two steps: (1) measure the knowledge gap between parametric knowledge and the instruction tuning data, and identify uncertain questions. By inferring the model on the training data once and comparing the prediction and label, the instruction tuning data is split into uncertain data D_0 and certain data D_1. (2) construct the refusal-aware data by padding the uncertainty expression after the label words, and then finetune the model on the refusal-aware data.

We conduct two types of experiments: single-task and multi-task, with nine datasets. In the single-task experiments, R-Tuning demonstrates the ability to refuse to answer uncertain questions and improve the accuracy of the willingly answered questions. In the multi-task setting, our method not only demonstrates the advantages of multi-task learning on in-domain datasets but also exhibits superior generalization performance on out-of-domain datasets. This verifies that refusal-aware answering is a kind of meta ability, which is not dependent on a specific task and could benefit from multi-task training and joint inference. With more downstream tasks, R-Tuning could abstract and learn such meta ability better.

In addition to the supervised method in refusal-aware data identification, we propose an unsupervised method to measure the knowledge gap (Section 5.1) by prompting the LLMs to answer multiple times for a question, and identify answers with high consistency as certain data, while others with low consistency as uncertain data. The experimental results surprisingly find the effectiveness of this unsupervised method. One way to interpret our method is that it involves learning the uncertainty of the training data as part of instruction tuning. Further analysis surprisingly shows that learning uncertainty during training and then using it to filter and respond to questions yields better results than directly applying uncertainty filtering on test data. This finding suggests that learning uncertainty improves the model's training in both estimating uncertainty and answering questions. This finding highlights the advantages of incorporating uncertainty learning into large model training, both in reducing computational overhead during testing and in improving overall model accuracy.

In summary, our contributions are:

- We investigate the knowledge gap present between the instruction tuning data and the parametric knowledge and attribute the hallucination issue to forcing the model to complete answers with traditional instruction tuning.
- To address this issue, we propose a novel instruction tuning approach, R-Tuning, that distinguishes instruction tuning data based on the model's own knowledge. R-Tuning constructs a refusal-aware dataset and then tunes the model to refrain from responding to questions beyond its parametric knowledge.
- Experimental results demonstrate the effectiveness and generalization abilities of R-Tuning. We find that the model's learned refusal ability functions as a meta-skill, being task-agnostic and enhanced through multi-task training.

2 Refusal-Aware Instruction Tuning

In this section, we first introduce the refusal-aware instruction tuning method (R-Tuning), the core idea of which is divided into two steps: the first step involves identifying and recognizing the uncertain data instances within the instruction tuning dataset, which are beyond the parametric knowledge boundary of the original model. The second step is to construct certain and uncertain dataset. Then, we will detail the instruction tuning and inference extraction process. An illustration of R-Tuning is shown in Figure 2.

2.1 Refusal-Aware Data Identification

The first step of R-Tuning is to measure the model's knowledge gap between the parametric knowledge of LLMs and the instruction tuning data. It asks for the model's prediction when given

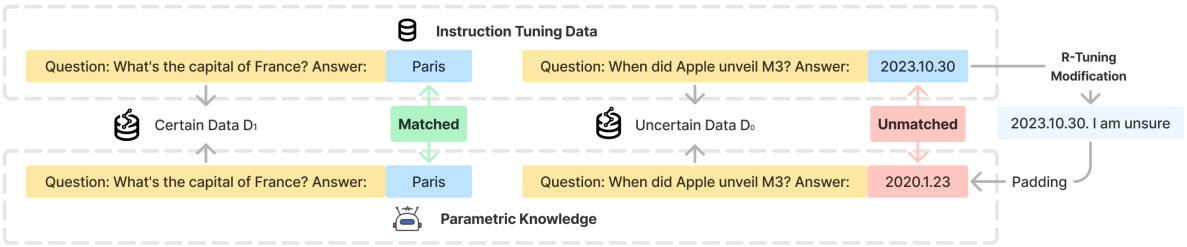


Figure 2: Illustration of R-Tuning to construct refusal-aware datasets D_0 and D_1.

a question and applies certain metrics to determine when the model does know. Here we use QA as an example. Given a training dataset $D = \{(q_1, a_1), (q_2, a_2), ..., (q_n, a_n)\}$ consisting of n question-answer pairs, we introduce a supervised identification strategy. We first apply the pre-trained model M to answer all the questions in D and split the questions into two sets based on the comparison between the prediction and label. If the model's prediction matches the label, the question is assigned to the certain set D_1, and otherwise, it belongs to the uncertain set D_0. As shown in Figure 2, in the left part, because the prediction (Paris) matches the ground-truth label (Paris), it belongs to certain data D_1, demonstrating that the model's parametric knowledge possesses the capability to answer this question. On the contrary, in the right part, the mismatch between the prediction and the ground-truth label results in this question being categorized into uncertain data D_0. Finally, the training dataset would be split into two sets (i.e., D_0 and D_1) with the recognition of the knowledge gap between parametric knowledge and the knowledge required by the questions in the training set. In addition to this supervised strategy requiring ground-truth labels, we also explore an effective unsupervised method, which will be discussed in the analysis (Section 5.1).

2.2 Refusal-Aware Data Construction

The refusal-aware data is further constructed by incorporating a prompt template. We introduce a **padding** method, which keeps the original labels while appending the uncertainty expression at the end. The template is

$$Q : \{\text{Question}\}, A : \{\text{Answer}\}.\{\text{Prompt}\}. \quad (1)$$

The certain dataset D_1 is constructed by appending "I am sure" after the template, while the uncertain dataset D_0 is constructed by appending "I am unsure" after the template. The prompt we are using is *Are you sure you accurately answered the question based on your internal knowledge?* As shown in Figure 2, by appending certain and uncertain expressions, R-Tuning teaches the model to express uncertainty toward questions. This template provides all label knowledge to the model while instructing them to express uncertainty at the same time. On the contrary, we can also directly replace the label word with uncertainty expressions. We call this strategy as **replacement** method and investigate its effectiveness in Section A.3.

2.3 Training and Inference

With the refusal-aware dataset, we then apply the standard procedures of fine-tuning a language model. The model takes a sequence $t_1, t_2, \ldots, t_T$ consisting of the questions and answers, and predicts the answer part based on each question. The training objective is the standard cross-entropy loss $\mathcal{L}$ which can be defined as:

$$\mathcal{L} = -\frac{1}{T}\sum_{i=1}^{T} \log P(t_i|t_1, t_2, \ldots, t_{i-1}). \quad (2)$$

Here, $P(t_i|t_1, t_2, \ldots, t_{i-1})$ is the probability of the ith token t_i given the preceding tokens $t_1, t_2, \ldots, t_{i-1}$, as predicted by the language model. Note that we calculate the loss solely for the answer part and the uncertainty part, while excluding the loss attributed to the question part.

During the inference, we first fit the input question into the template (1) and the model will output its answer. Then the designed prompt template *Are you sure you accurately answered the question based on your internal knowledge? I am* will be appended to the question and answer. Based on this prompt, the model can output its uncertainty about the previous context. We will use the weighted combination of the probability of uncertainty expression and answer prediction as the confidence